Exercises

Exercise 1.
Check the linearity and time-invariance of the discrete system described by equation $y[n] = x[n] + 2$.

Exercise 2.
The output of a linear time-invariant discrete system to the input signal $x[n] = u[n]$ is $y[n] = 2^{-n}u[n]$. Find the impulse response $h[n]$. Is the system stable ?

Exercise 3.
Find the convolution $y[n] = x[n] * x[n]$ for $x[n] = u[n] - u[n-5]$.

Exercise 4.
Find the convolution of discrete-time signals $x[n] = e^{-|n|}$ and $h[n] = u[n+5] - u[n-6]$.

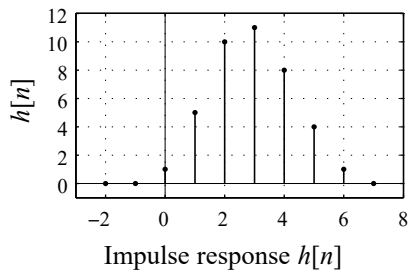
Exercise 5. A discrete system consists of systems with impulse responses $h_1[n] = e^{-an}u[n]$, $h_2[n] = e^{-bn}u[n]$, and $h_3[n] = u[n]$. Find the impulse response of the resulting system for:

(a) Systems $h_1[n]$, $h_2[n]$, and $h_3[n]$ connected in parallel,
(b) System $h_1[n]$ connected in parallel with a cascade of systems $h_2[n]$ and $h_3[n]$.

Exercise 6. Consider three causal linear time-invariant systems in cascade. Impulse responses of these systems are $h_1[n]$, $h_2[n]$, and $h_2[n]$, respectively. The impulse response of the second and the third system is $h_2[n] = u[n] - u[n-2]$, while the impulse response of the whole system,

$$h[n] = h_1[n] * h_2[n] * h_2[n],$$

is shown in Figure below.



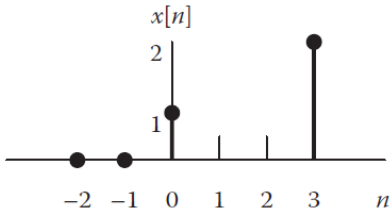
Impulse response $h[n]$

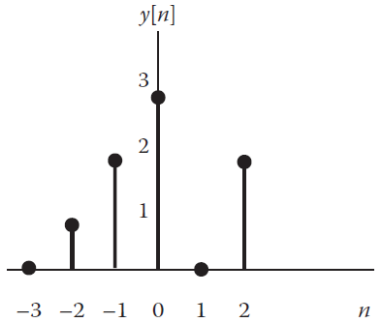
Find $h_1[n]$ and $y[n] = h[n] * x[n]$, with $x[n] = \delta[n] - \delta[n-1]$.

Exercise 7

(a) If $r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases}$, called a unit ramp sequence, find $y[n] = 2r[1-n]$

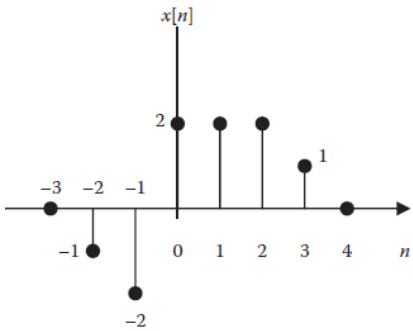
(b) A general way to express any discrete signal is given as $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$
Write down expressions for the sequences shown in the Figures below





(c) Given the discrete-time signal in the figure below, sketch the following signals:

1. $y[n] = x[n-3]$
2. $z[n] = x[n] - x[n-1]$



Exercise 8

Determine whether the following systems are linear

$T_1(x[n]) = x[n]\, u[n]$

$T_2(x[n]) = x[n] + n\, x[n+1]$

$T_3(x[n]) = 2x[n-2] + 5$

$T_4(x[n]) = \sum_{k=-\infty}^{n+5} 2x[k]$

$T_5(x[n]) = 3x^2[n]$

Exercise 9

Determine whether the following linear systems are time-invariant.

a) $y[n] = L(x[n]) = 10 \sin(0.1\pi n)\, x[n]$

b) $y[n] = L(x[n]) = x[n+1] - x[1-n]$

c) $y[n] = L(x[n]) = 14x[n] + 12x[n-1] + 14x[n-2]$

Exercise 10

Given the difference equation: $y[n] = 0.25\, y[n-1] + x[n]$, identify the nonzero system coefficients.

Exercise 11

Assume we have a linear time-invariant system

$$y[n] = 0.5\, x[n] + 0.25\, x[n-1]$$

with an initial condition $x[-1] = 0$.

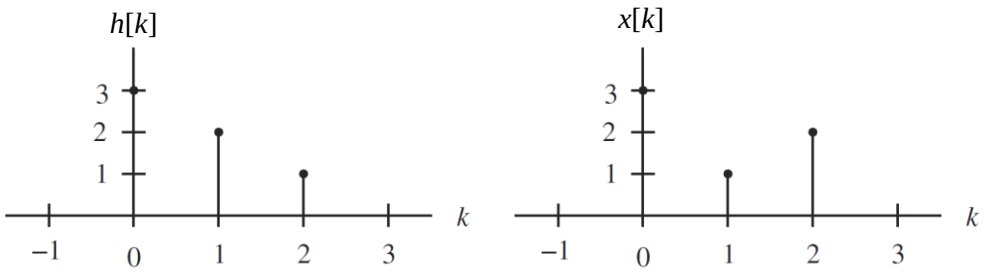
a. Determine the unit-impulse response $h[n]$.
b. Draw the system block diagram.
c. Write the output using the obtained impulse response.

Exercise 12

Using the sequences defined in the figure below, evaluate the digital convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

a. by the graphical method.
b. by applying the formula directly.



Exercise 13

Determine the stability of the following linear system.

$$y[n] = 0.5\, x[n] + 100\, x[n-2] + 20\, x[n-10]$$

Exercise 14

Determine whether the following signals are energy signals, power signals, or neither.

(a) $x(t) = \begin{cases} e^{-at}, & 0 < t < \infty,\ a > 0 \\ 0, & \text{otherwise} \end{cases}$

(b) $x(t) = A\cos(\omega t + \theta)$

(c) $x[n] = 10e^{j2n}$

(d) $x(t) = \begin{cases} t, & 0 < t < \infty \\ 0, & \text{otherwise} \end{cases}$

(e) $x(t) = e^{-3t}u(t)$

(f) $x(t) = (e^{-2t} + 1)u(t)$

(g) $x(t) = t^{-1/2}u(t-2)$